Lecture 10

Cauchy's formula and rigidity

Objectives. *Cauchy's formula recovers a holomorphic function inside a disk from its values on the boundary. We derive its power series and the estimates on its derivatives. These results also justify limits and integrals that define holomorphic functions.*

By the end of this lecture you will be able to:

- Recover values and derivatives inside a disk by Cauchy's formula.
- Apply the identity theorem and maximum modulus principle on a domain.
- Prove Liouville's theorem and the fundamental theorem of algebra.
- Check domination before integrating a holomorphic parameter.

10.1 Cauchy's integral formula on a disk

Recall that $D(a,r) = \{z : |z-a| < r\}$. Its boundary is traversed counter-clockwise. The closed disk must lie inside the open set of holomorphy. An open disk alone does not contain its boundary.

Lecture 9 constructed a single primitive on a star-shaped domain. A region with a hole need not have such a primitive. Instead we use a primitive on each small disk and add the resulting boundary integrals. This proves a statement about the sum of all boundary components.

We first extend the local result from Lecture 9 to the boundaries used here. A region in the following lemma may have holes. Positive orientation keeps the region on the left, so a hole is traversed clockwise. Here a region means an open set lying locally on exactly one side of each smooth boundary arc.

Lemma 10.1 (Boundary cancellation)**.** *Let E be a bounded region whose boundary consists of finitely many simple closed curves made from line segments and circular arcs. The curves are disjoint. If f is holomorphic on an open neighbourhood of $\overline{E}$, then*

$$\int_{\partial E} f(\zeta)\,\mathrm{d}\zeta = 0.$$

The integral denotes the sum over all oriented boundary components.

Proof. By Theorem 9.4, a holomorphic function has a primitive on every disk contained in its open set of holomorphy. Cover the compact set $\overline{E}$ by smaller disks whose concentric larger disks still lie in that open set. A sufficiently fine square grid has the property that each square meeting $\overline{E}$ lies in one larger disk. To see this, choose a finite subcover by the smaller disks and make each square's diameter smaller than the smallest gap to the corresponding larger circle.

Choose a translated grid with no boundary vertex on a grid line and no grid line tangent to a boundary circle or coincident with a boundary segment. Only finitely many squares meet $\overline{E}$. Each boundary arc meets their grid lines finitely many times. Cutting E along the grid therefore gives finitely many pieces inside disks with primitives. Orient each boundary edge with its piece on the left. At each vertex, every local sector belonging to the piece has one entering and one leaving edge. The numbers of incoming and outgoing edges therefore agree. These finitely many directed edges decompose into closed walks: follow unused edges until a vertex repeats, remove the resulting closed walk, and repeat. Removing a closed walk preserves the balance at every vertex, so the process accounts for every edge.

Each walk lies in one disk. Its integral is zero by Equation (9.2), which evaluates a primitive at the endpoints. Add these equalities over all pieces. On an original boundary arc the piece lies on the same side as E, so that arc inherits E's orientation. Every internal grid edge separates two pieces on opposite sides and occurs twice with opposite directions. Only the oriented arcs of ∂E remain. ∎

The grid in the figure separates the global boundary from the small pieces on which a primitive is available.

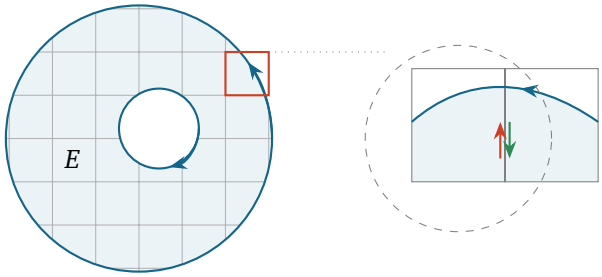


Figure 10.1: A region with a hole is cut by a square grid. The outer boundary is counter-clockwise and the hole boundary is clockwise. The enlarged boundary cells show a curved original boundary and opposite arrows on a shared grid edge. A dashed disk indicates a local primitive neighbourhood.

The lemma uses local primitives and cancellation. It makes no assertion about the integral around a hole alone. For example, on an annulus the integrals of $1/\zeta$ around its outer and inner boundaries cancel.

Theorem 10.2 (Cauchy's integral formula)**.** *If f is holomorphic on an open neighbourhood of $\overline{D(a,r)}$, then, for every $z \in D(a,r)$,*

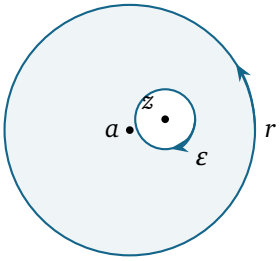
$$f(z) = \frac{1}{2\pi \mathrm{i}} \oint_{\partial D(a,r)} \frac{f(\zeta)}{\zeta - z}\, \mathrm{d}\zeta. \tag{10.1}$$

Proof. Remove a closed disk $\overline{D(z,\varepsilon)}$ contained in $D(a,r)$. The function $f(\zeta)/(\zeta - z)$ is holomorphic on an open neighbourhood of the remaining closed region. Boundary cancellation gives equal integrals on the two circles when both are oriented counter-clockwise. Parametrising the small circle gives

$$\oint_{\partial D(z,\varepsilon)} \frac{f(\zeta)}{\zeta - z}\, \mathrm{d}\zeta = \mathrm{i} \int_0^{2\pi} f(z + \varepsilon e^{\mathrm{i}t})\, \mathrm{d}t \longrightarrow 2\pi \mathrm{i} f(z).$$

The convergence follows from continuity at z, uniformly on the small circle. The large-circle integral is independent of ε. ∎

The small hole in the proof removes the singularity of the kernel.



The region has counter-clockwise outer boundary and clockwise inner boundary. Moving the inner integral to the other side reverses it.

10.2 Power series and derivative estimates

A function is *analytic* if near each point it is the sum of a convergent power series. Lecture 9 proved that analytic functions are holomorphic. The converse follows by expanding the Cauchy kernel.

Theorem 10.3. *Let f be holomorphic on an open set $\Omega \subset \mathbb{C}$ and let $a \in \Omega$. The function has derivatives of every order, and*

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(z-a)^n, \qquad |z-a| < \operatorname{dist}(a, \mathbb{C} \setminus \Omega).$$

For $\Omega = \mathbb{C}$ the distance is interpreted as infinity. If $\overline{D(a,r)} \subset \Omega$, then

$$f^{(n)}(z) = \frac{n!}{2\pi i} \oint_{\partial D(a,r)} \frac{f(\zeta)}{(\zeta - z)^{n+1}}\, d\zeta, \qquad z \in D(a,r),\ n \geq 0. \tag{10.2}$$

Writing $M_r = \max_{|\zeta - a| = r} |f(\zeta)|$, we have

$$\left| f^{(n)}(a) \right| \leq \frac{n! M_r}{r^n}, \qquad n \geq 0. \tag{10.3}$$

Proof. For $|z-a| < r$ and $|\zeta - a| = r$, the geometric series gives

$$\frac{1}{\zeta - z} = \sum_{n=0}^{\infty} \frac{(z-a)^n}{(\zeta - a)^{n+1}}.$$

On each smaller disk $|z-a| \leq r_0 < r$, the ratio has modulus at most $r_0/r < 1$. The convergence is uniform in both variables there. Cauchy's formula and uniform integration, Theorem 1.2, give a power series with coefficients

$$a_n = \frac{1}{2\pi i} \oint_{\partial D(a,r)} \frac{f(\zeta)}{(\zeta - a)^{n+1}}\, d\zeta.$$

By Theorem 1.5, power series differentiate term by term on smaller disks. Hence $f^{(n)}(a) = n! a_n$ and f has derivatives of all orders near a. The same argument at every point proves smoothness throughout Ω. The coefficients are independent of r, since they are the derivatives at a. Taking all admissible r proves the claimed radius bound.

To obtain the derivative formula at an arbitrary z inside the fixed circle, differentiate its integral. For the first derivative the kernel quotient is

$$\frac{1}{h}\left(\frac{1}{\zeta - z - h} - \frac{1}{\zeta - z}\right) = \frac{1}{(\zeta - z - h)(\zeta - z)}.$$

The distance from z to the circle is positive. Thus this quotient converges uniformly on the circle to $(\zeta - z)^{-2}$. For higher derivatives, let $m \geq 1$ be an integer and write $A = \zeta - z$. For sufficiently small $h \neq 0$, factoring the difference of two mth powers gives

$$\frac{(A-h)^{-m} - A^{-m}}{h} = \sum_{j=0}^{m-1} \frac{1}{(A-h)^{j+1} A^{m-j}}.$$

The positive distance to the circle bounds $|A|$ away from zero. For sufficiently small h, it also bounds $|A - h|$ away from zero. Each term in this finite sum therefore converges uniformly to A^{-m-1}. The quotient converges uniformly to mA^{-m-1}. Starting with Cauchy's formula, induction now differentiates the integral and multiplies its coefficient successively by $1, 2, \ldots, n$. This proves the derivative formula. Finally the ML-estimate, Equation (9.1), gives $n! M_r (2\pi r)/(2\pi r^{n+1})$, as required. ∎

For example, Cauchy's derivative formula evaluates an integral without parametrising the circle:

$$\oint_{|\zeta|=1} \frac{e^{2\zeta}}{\zeta^4}\,\mathrm{d}\zeta = \frac{2\pi \mathrm{i}}{3!}\,\frac{\mathrm{d}^3}{\mathrm{d}z^3} e^{2z}\Big|_{z=0} = \frac{8\pi \mathrm{i}}{3}.$$

The denominator's fourth power selects the third derivative.

10.3 Liouville and the fundamental theorem of algebra

An *entire* function is holomorphic on all of $\mathbb{C}$.

Theorem 10.4 (Liouville). *Every bounded entire function is constant.*

Proof. If $|f| \leq M$ on $\mathbb{C}$, Cauchy's estimate gives $|f'(a)| \leq M/r$ for every $r > 0$. Letting $r \to \infty$ shows $f'(a) = 0$ at every $a \in \mathbb{C}$. Restricting f to a segment and applying the real fundamental theorem of calculus shows that its endpoint values agree. Thus f is constant. ∎

Theorem 10.5 (Fundamental theorem of algebra). *Every complex polynomial of degree $n \geq 1$ has exactly n roots, counted with multiplicity.*

Proof. Let $p(z) = b_n z^n + \cdots + b_0$ with $b_n \neq 0$. As $|z| \to \infty$, the quotient $p(z)/z^n$ tends to b_n. For sufficiently large $|z|$, this gives $|p(z)| \geq |b_n|\,|z|^n/2$. If p had no root, its continuous modulus would also have a positive minimum on the remaining compact disk. Consequently $1/p$ would be bounded and entire. Liouville would make $1/p$, and hence p, constant, a contradiction.

A root a gives a factor $z - a$ by polynomial division. Repeating the existence argument on the quotient gives n linear factors, with a nonzero constant left over. No further root is possible in that product. The number of repeated factors at a root is its multiplicity. ∎

10.4 Identity and maximum modulus

The first nonzero coefficient determines a zero locally. If $f(a) = 0$ and its Taylor series is not identically zero, let $m \geq 1$ be the first index with nonzero coefficient. Factoring the series gives

$$f(z) = (z - a)^m g(z), \qquad g(a) \neq 0,$$

where g is holomorphic near a. By continuity, g stays nonzero on a smaller disk. The zero a is therefore isolated and has *order* m.

Theorem 10.6 (Identity theorem). *Let f and g be holomorphic on a domain Ω. If they agree on a sequence of distinct points converging to a point of Ω, then $f = g$ throughout Ω.*

Proof. Apply the preceding factorisation to $f - g$ at the limit point. Isolated zeros cannot accumulate there, so all Taylor coefficients vanish and $f - g$ vanishes on a disk. Let U be the set of points where $f - g$ vanishes on a neighbourhood. This set is nonempty and open. It is also closed relative to Ω: a point in its relative closure either belongs to U already or is a limit of distinct zeros, to which the same Taylor argument applies. Connectedness means that Ω cannot be split into disjoint nonempty relatively open sets. If U were a proper subset, it and its complement would give such a split, since U is both open and relatively closed. Thus $U = \Omega$. ∎

An *analytic continuation* of f to a larger domain is a holomorphic function there agreeing with f on its original domain. The theorem proves uniqueness on a fixed larger domain. It does not assert existence.

Theorem 10.7 (Maximum modulus principle). *If a holomorphic function on a domain has a local maximum of its modulus at an interior point, it is constant.*

Proof. Suppose the maximum occurs at a. If $f(a) = 0$, the local bound forces $f = 0$ on a disk, and the identity theorem finishes the proof. Otherwise multiply f by a constant of modulus one so that its value at a is the positive number $M = |f(a)|$. On each sufficiently small circle about a, Cauchy's formula at the centre gives

$$M = \frac{1}{2\pi} \int_0^{2\pi} f(a + re^{it})\, \mathrm{d}t.$$

On this circle, $\operatorname{Re} f \leq |f| \leq M$. The continuous nonnegative function $M - \operatorname{Re} f$ has zero integral, so it is zero everywhere on the circle. The modulus bound then forces the imaginary part to vanish. Thus $f = M$ on every such circle and on the enclosed disk. The identity theorem extends the equality to the domain. ∎

In particular, if f is holomorphic on a neighbourhood of a closed disk, its maximum modulus on the disk is attained on the boundary. Otherwise an interior maximiser would make it constant, in which case the boundary also attains the maximum.

10.5 Harmonic real and imaginary parts

A real-valued C^2 function u on an open subset of $\mathbb{R}^2$ is *harmonic* if its Laplacian vanishes:

$$\Delta u = u_{xx} + u_{yy} = 0.$$

If $f = u + \mathrm{i}v$ is holomorphic, the Taylor theorem makes u and v smooth: $\partial_x f = f'$ and $\partial_y f = \mathrm{i}f'$ by Proposition 9.2, and induction applies since each complex derivative is holomorphic. Cauchy–Riemann, Proposition 9.2, now gives

$$u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0, \qquad v_{xx} + v_{yy} = -u_{yx} + u_{xy} = 0.$$

The mixed partials agree because they are continuous. This deduction would not follow merely from the existence of first partial derivatives.

For example, $\operatorname{Re}(z^2) = x^2 - y^2$ is harmonic, since its second partials are 2 and -2. The real part of e^z is $e^x \cos y$. Its second partials cancel as well. In a steady heat problem these are local temperature profiles with no source term. A prescribed boundary temperature is additional data, not a consequence of harmonicity.

10.6 Morera, limits and parameter integrals

Theorem 10.8 (Morera). *Let $f : \Omega \to \mathbb{C}$ be continuous on an open set. If the boundary integral of f is zero on every filled triangle contained in Ω, then f is holomorphic.*

Proof. Work in a disk inside Ω. The segment construction in the proof of Theorem 9.4 uses only continuity and zero triangle integrals. It therefore gives a primitive F with $F' = f$ on the disk. Since F is holomorphic, Theorem 10.3 gives a second complex derivative. Hence $f = F'$ is holomorphic. Such disks cover Ω. ∎

If holomorphic functions f_n converge uniformly on every compact subset of Ω, their limit f is holomorphic. Indeed, the limit is continuous on each compact disk, and the ML-estimate passes every triangle integral to the limit. Goursat and Morera apply. Their derivatives also converge locally uniformly: cover a compact set by finitely many smaller disks and use Cauchy's estimate for $f_n - f$ on the corresponding larger disks.

Theorem 10.9 (Holomorphic parameter integrals). *Let $E \subset \mathbb{R}^d$ be measurable and $\Omega \subset \mathbb{C}$ open. Suppose $h(t, z)$ is measurable in t for each z and holomorphic in z for every t outside one fixed null set. For every compact set $K \subset \Omega$, suppose there is a nonnegative $g_K \in L^1(E)$ such that $|h(t, z)| \le g_K(t)$ for all $z \in K$, outside one null set that may depend on K. Then the function*

$$H(z) = \int_E h(t, z)\, \mathrm{d}t$$

is holomorphic on Ω. Its derivatives satisfy

$$H^{(n)}(z) = \int_E \partial_z^n h(t, z)\, \mathrm{d}t, \qquad n \ge 0. \tag{10.4}$$

Proof. First redefine $h(t, z) = 0$ for every z when t lies in the fixed exceptional null set of the holomorphy assumption. Every integral is unchanged, and $h(t, \cdot)$ is now continuous for every t. The domination on compact disks and dominated convergence (Theorem 3.5) make H continuous. Fix a triangle whose filled interior lies in Ω. On its boundary, domination by g_K times the finite speed of each linear parametrisation is integrable over E times the parameter interval. Product measurability follows by approximating the continuous parameter dependence by its values on finer partitions. Each approximant is measurable in

the integration variable and stepwise constant in the curve parameter. The pointwise limit is therefore measurable. Fubini, the standing fact M4 in Theorem 4.1, therefore gives

$$\oint_{\partial T} H(z)\,\mathrm{d}z = \int_E \left(\oint_{\partial T} h(t,z)\,\mathrm{d}z \right) \mathrm{d}t = 0.$$

The inner integral vanishes by Goursat outside the fixed null set. Morera proves holomorphy. Fubini applies to real and imaginary parts.

Fix z and a closed disk $\overline{D(z,r)} \subset \Omega$. Cauchy's estimate gives $\left|\partial_z^n h(t,z)\right| \le n! g_K(t)/r^n$ with $K = \overline{D(z,r)}$. The derivatives are measurable in t, since they are limits of difference quotients. Apply Cauchy's derivative formula to H and exchange its circle integral with the t integral by Fubini and the same domination. The inner circle integral is $\partial_z^n h(t,z)$ by Cauchy's formula. This proves the stated identity and its absolute integrability. ∎

For a complex Gaussian, let $\operatorname{Re} a > 0$. On each compact parameter set we have $\operatorname{Re} a \ge c > 0$, so $e^{-cx^2/2}$ dominates $e^{-ax^2/2}$ in modulus on $\mathbb{R}$. The parameter theorem makes the integral holomorphic in a. The real Gaussian formula of Theorem 4.3 agrees on $a > 0$ with the holomorphic function $\sqrt{2\pi}/\sqrt{a}$ on this half-plane. Here $\sqrt{a} = \exp((\log a)/2)$ uses the principal logarithm of § 9.2. This definition makes the root holomorphic and nonzero on the half-plane. The identity theorem therefore gives

$$\int_{\mathbb{R}} e^{-ax^2/2}\,\mathrm{d}x = \sqrt{\frac{2\pi}{a}}, \qquad \operatorname{Re} a > 0. \tag{10.5}$$

Here the right side equals $\sqrt{2\pi}/\sqrt{a}$, since $-\pi/2 < \arg a < \pi/2$. Lecture 13 applies the same parameter theorem to Euler's Gamma integral.

Summary.

- Cauchy's formula requires a closed disk inside an open holomorphic set.
- A holomorphic function is analytic. Cauchy's estimates bound all derivatives by boundary values.
- Bounded entire functions are constant. Every nonconstant complex polynomial factors into linear factors.
- On a domain, accumulating interior agreement gives identity, and an interior local maximum of the modulus forces constancy.
- Compact-set integrable domination makes holomorphic parameter integrals holomorphic and permits all complex derivatives under the integral.

Lecture 11 uses the boundary formula on annuli and around isolated singularities.

References

E. M. Stein and R. Shakarchi, *Complex Analysis*, Princeton, 2003, Chapter 2, §§2, 4–5, pp. 37–41, 45–54. The boundary-cancellation proof above spells out the passage from local primitives for the finite line-and-circle contours used in this course.